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Software Development Individual Project

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CS -40

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CS - 40

Declaration

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Abstract

Sleep is essential to maintaining health, affecting physical and mental well-being. But while most people hope to get a better night's sleep, many do not know how or why much less what it might take for them to improve theirs. An application that is data driven which will help in tracking, analyzing, and improving sleep by gathering easily accessible real-time data from wearable devices using machine learning algorithms.

The Application integrates with core sleep metrics, HRV, and activity data from popular wearable devices such as Fitbit & Apple Watch. It also includes user-reported data like sleep journals and medication use in order to provide personalized suggestions for better sleep hygiene. It uses predictive analytics using linear regression, to spot the trends and provocations, thus helping the users through predictable practitioners.

Keyword.

- Wearable devices
- Machine learning
- Random forrest regression
- Health Data analysis
- Predictive analytics
- Caffeine intake
- Service tier
- Business tier
- Data access tier
- Prototyping
- Brainstorming session
- Usability
- Scalability
- machine learning model
- Surveys/questionnaires
- Prototyping
- User authentication

Acknowledgment

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Abbreviations table

1. **AI** - Artificial Intelligence
2. **HRV** - Heart Rate Variability
3. **ML** - Machine Learning
4. **SRS** - System Requirements Specification
5. **API** - Application Programming Interface
6. **SLEP** - Social, Legal, Ethical, and Professional
7. **GDPR** - General Data Protection Regulation
8. **HIPAA** - Health Insurance Portability and Accountability Act
9. **CCPA** - California Consumer Privacy Act
10. **UI** - User Interface
11. **DFD** - Data Flow Diagram
12. **FR** - Functional Requirement
13. **NFR** - Non-Functional Requirement
14. **CC BY-SA** - Creative Commons Attribution-ShareAlike

Chapter 4: System Requirements Specification (SRS)

4.1 Chapter Overview

This section investigates the functional and non-functional requirement aspects of the AI-enhanced sleep quality predicting web application targeted at university students. The paper were elicited, and evaluates the findings of this analysis, reaching a detailed comprehension of the system's overall architecture and its components' behavior.

4.2 Stakeholder Analysis

4.2.1 Onion Model

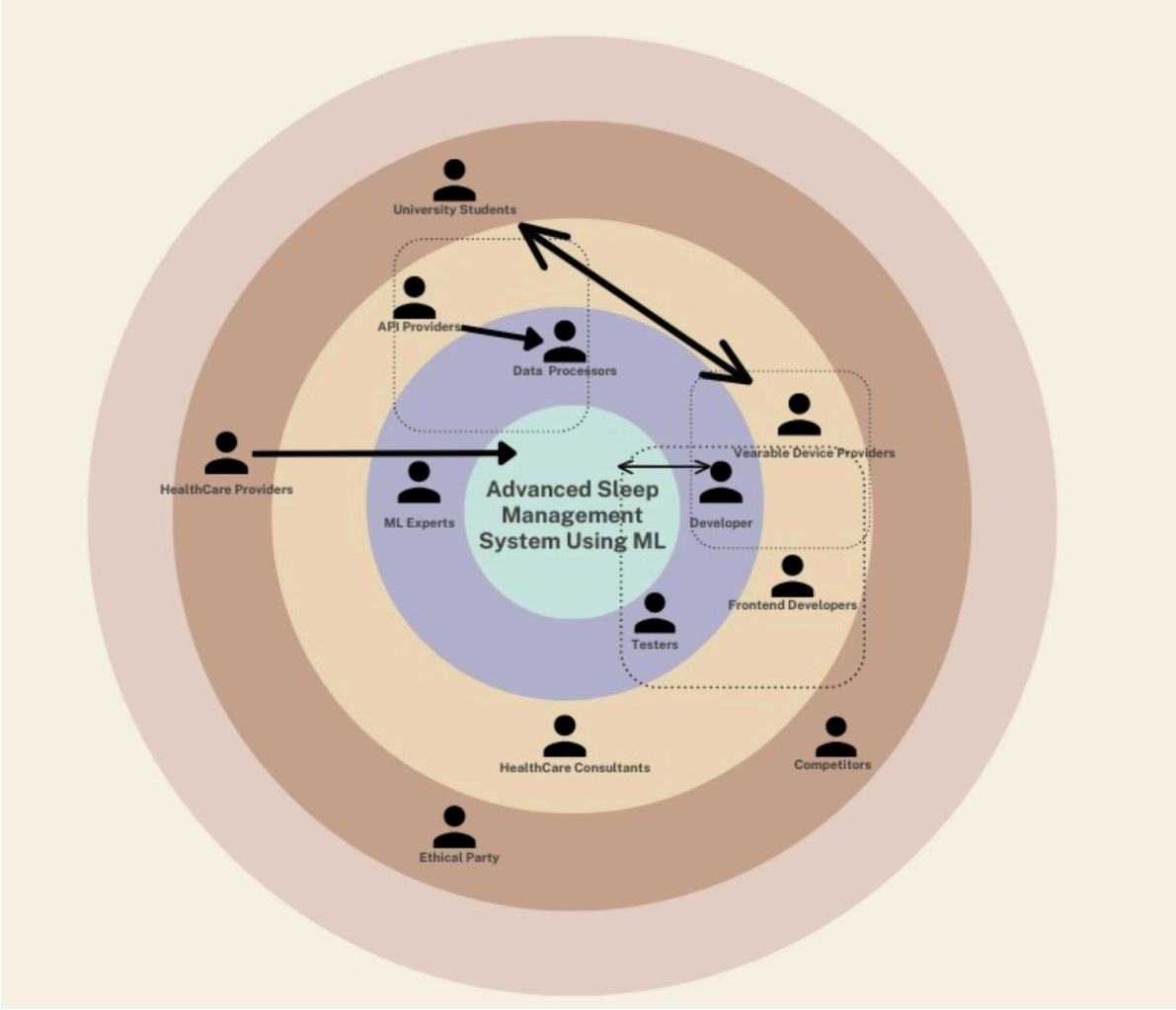


Figure 1: Onion Model

4.2.2 Stakeholder Descriptions

Stakeholder	Viewpoint

Functional beneficiary	
University Students	They directly use the application for get sleep quality prediction and recommendation.
Financial beneficiary	
API providers	Earn some income from API integration
Wearable device providers	Earn some income by selling devices
Social beneficiary	
HealthCare consultants	Can use the information for the betterment of public health policies
Operational beneficiary	
Developers	Build, maintain, and store the data of the system
Negative Stakeholders	
Competitors	Have a threat to the app's capability then losing the market share.
Regulatory	
None	
Experts	
ML Experts	Explain about the what is the best regression. And give advice to improve the model accuracy.
Health Providers,researchers	Giving a professional knowledge about sleep quality.
Neighboring systems	

4.3 Selection of Requirement Elicitation Techniques/Methods

The process of acquiring relevant project needs using a variety of instruments or methodologies is known as requirement elicitation. This section discusses a few of these options , as well as whythey were chosen for use in this project. Literature reviews, prototypeing and formal interviews are among the subject explored.

The AI-driven sleep quality prediction web app need to gather details information from stakeholders(university students,doctors, technical experts) the goals is to understand the users' needs for the app as well as gather the correct data and develop the precess.

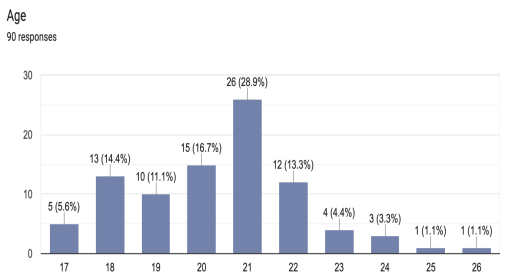
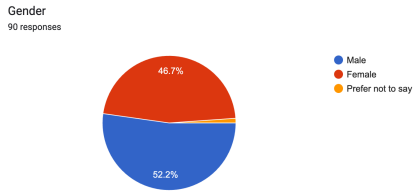
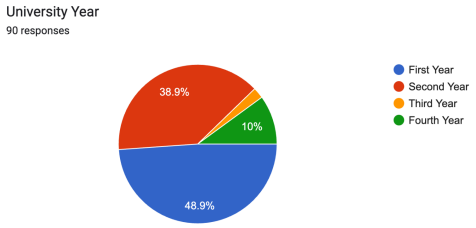
Method 1: Surveys/Questionnaires
● **Purpose**: Collect a large amount of university students about their sleep habits, study habits, and physical activity. ● **Methods**: send the Google form to get the anonymous students' data.(*Sleepless Nights? Let's Fix That!*, 05/09/2024) ● **Benefit**: Quick and low-cost method for obtaining structured data from big audiences.
Method 2: Interviews
● **Purpose**: get motivation and professional advice on what we should improve our project ● **Method**: Meet a insomnia doctor or a research team. ● **Benefits**: get a deep knowledge of the topic.
Method 3: Prototyping
● **Purpose**: Evaluate how users interact with a web app design and features ● **Method**: Create a prototype using Figma and get student group feedback about the app. ● **Benefits**: get to know about usability issues.
Methods 4: Brainstorming Session

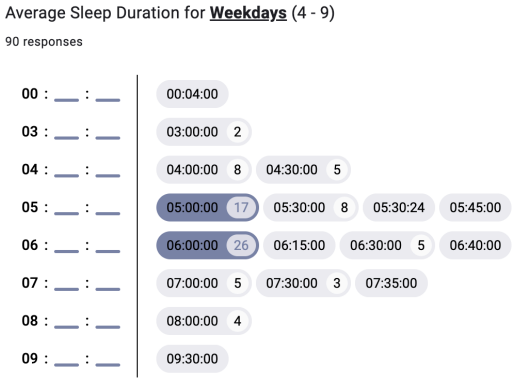
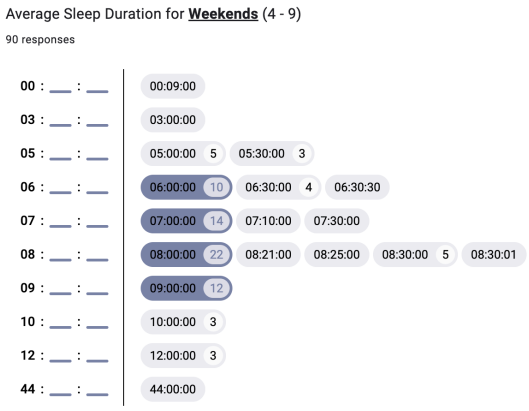
- **Purpose**: to generate new ideas and solutions using help from stakeholders.(experts, supervisors)
- **Method**: Organized brainstorming meetings will be conducted with key stakeholders.
- **Benefits**: it helps to improve the app features.

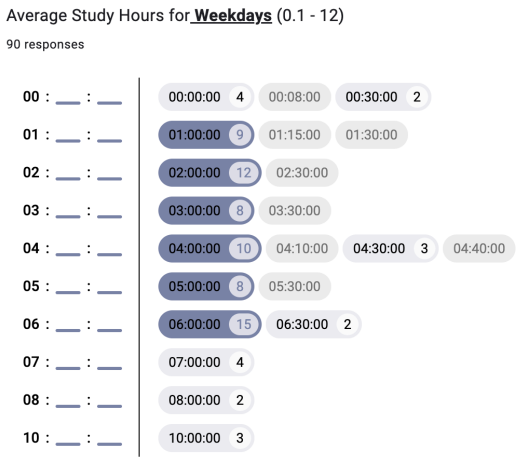
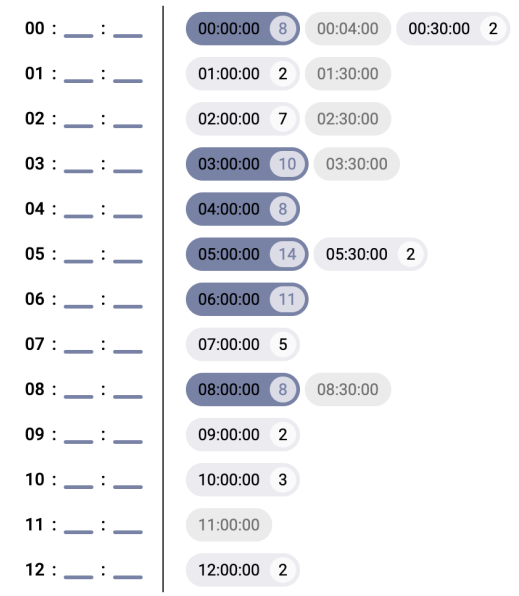
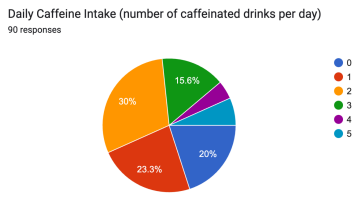
4.4 Discussion / Analysis of Results

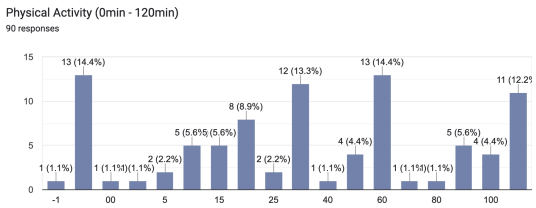
Method 1: **Survey/Questionnaire**
Get a comprehensive understanding of students' sleep habits, study patterns, and lifestyle factors. This structural data helps for statistical analysis, the relationship between sleep quality and factors like caffeine intake, physical activity, etc. According to the survey, most of the students do not get sufficient sleep because of long study hours and screen time. High caffeine consumption was found to be strongly associated with poor sleep quality. Additionally, this will help to train a model of Machine Learning. Only IIT students were included in the survey sample, which might have an impact on how broadly applicable the results are. To overcome this restriction, future iterations will try to increase the sample size. (*Sleepless Nights? Let's Fix That!*, 05/09/2024)
Method 2: **Interviews**
Interviews with insomnia doctors and research teams offered professional insights. This discussion depends on the people's understanding of this problem.
Method 3: **Prototyping**
user feedback on the prototype shows challenges. prototyping allows to early detection the design flaws and provides actionable insights. This helps to final project become more attractive for the user.

Question research

Question	Response	Analyse
Age		According this chart most of the students are in age of 21. Others are in age 17 to 26. According this all the students are in university year 1 to 4.
Gender		According this pie chart most of the students who fill this google form are male.
University Year		According this graph most of students who fill this chart are first year students.

Average sleep duaration	 	The length of sleep on weekdays and weekends is entered into the Google form. because the Google form's user finds it challenging to complete the seven days. Because weekends are free, the majority of individuals sleep properly on weekdays and properly on weekends. Average sleep hours = (average sleep hours weekdays + average sleep hours weekends) / 2

Average study hours	 	The length of study on weekdays and weekends is entered into the Google form. because the Google form's user finds it challenging to complete the seven days. But weekends are free, the majority of individuals sleep properly on weekdays and properly on weekends. Average study hours = (average study hours weekdays + average study hours weekends) / 2
Caffeine intake		According the this graph most of the students does not drink the caffeine.

Physical activity level		According the graph most students physical activity level is 60.

4.5 Use Case Diagram

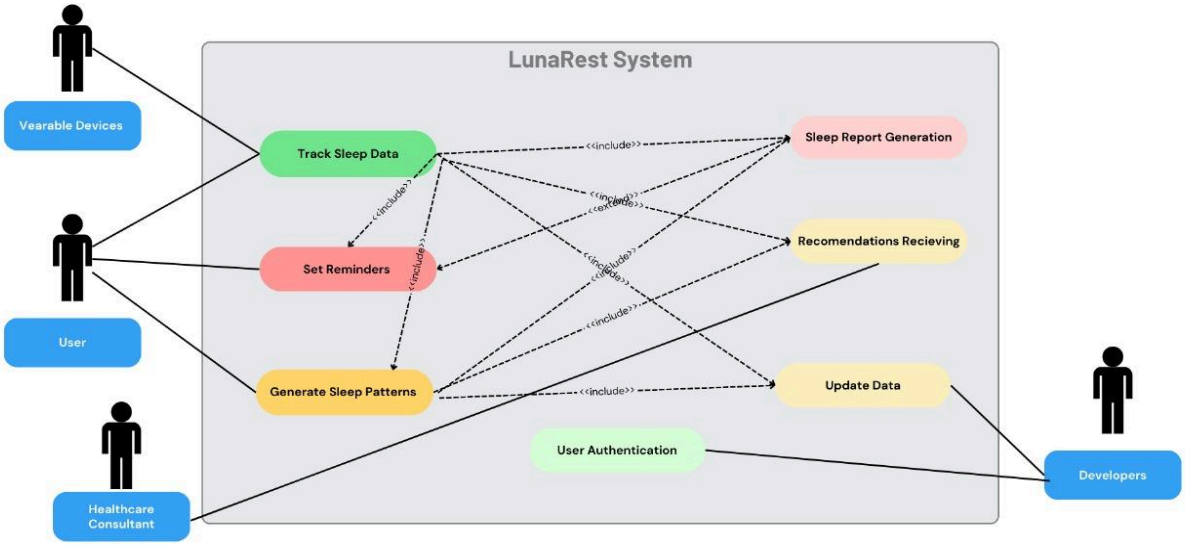


Figure2: LunaRest Diagram

4.6 Use Case Descriptions

Use Case Name	Track Sleep Data	
Use Case ID	UC-001	
Description	This can get input into the user's sleep average.	
Priority	High	
Primary Actor	User	
Supporting Actor	Wearable Devices	
Pre-Conditions	The user must log in to the system The user should give access to sleep data from wearable device	
Trigger	The user should activate the wearable device	
Main flow	**Actors**	**System**
	user	Authenticate the user
Exception flow	**Actors**	**System**
	user	If the wearable device is disconnected, give a notification to reconnect it.
Alternate flow	**Actors**	**System**
	-	-
Exclusions	Data from unsupported wearable devices.	
Post Conditions	Get the sleep data send it to the predict the sleep quality and stored this data	

Use Case Name	Set Reminders	
Use Case ID	UC-002	
Description	It allows user to personalized their bedtime, physical activity, and caffeine intake	
Priority	Medium	
Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	The user must log in to the system.	
Trigger	user should select the “Set Reminder” option	
Main flow	**Actors**	**System**
	User selects a reminder type like bedtime	Show the reminder customization and store reminders in the system.
Exception flow	**Actors**	**System**
	User input is invalid	Display an error.
Alternate flow	**Actors**	**System**
	The user requests reminder customization.	Updates reminder parameters based on user input.
Exclusions	Integration with calendar apps	
Post Conditions	Reminders are activated then and the notification to the user.	

Use Case Name	Sleep Report Generator	
Use Case ID	UC-003	
Description	The app generates the details of user sleep patterns as a graph.	
Priority	High	
Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	The user must be logged in to the system and input his data.	
Trigger	Request the sleep report	
Main flow	**Actors**	**System**
	User calls report generation	Processing the sleep data and generating it on the screen.
Exception flow	**Actors**	**System**
	User enter invalid data	Display a alert
Alternate flow	**Actors**	**System**
Exclusions	The user can not customize the graph template.	

Use Case Name	Sleep Report Generator
Post Conditions	Report is shown to the user.

Use Case Name	Recommendation Receiving	
Use Case ID	UC-004	
Description	The system provides recommendations to increase sleep quality and maintain the sleep quality	
Priority	High	
Primary Actor	User	
Supporting Actors	Healthcare consultant	
Pre-Conditions	The user should enter his data (sleep duration, study hours, etc.)	
Trigger	Always predict the sleep quality it shows the recommendations	
Main flow	**Actors**	**System**
	User requesting the recommendations.	Analyze the sleep quality and display recommendations.
Exception flow	**Actors**	**System**
	-	-
Alternate flow	**Actors**	**System**
	-	-

Use Case Name	Recommendation Receiving
Exclusions	Giving medical advice.
Post Conditions	The recommendations are saved in the database for the future.

Use Case Name	Generate the sleep patterns	
Use Case ID	UC-005	
Description	Analyze the user's long-term sleep patterns.	
Priority	High	
Primary Actor	User	
Supporting Actors	Developers	
Pre-Conditions	The user should enter his data.	
Trigger	The user should enter the “pattern analyse” option	
Main flow	**Actors**	**System**
	System generate the sleep patterns	Machine learning algorithms do that.
Exception flow	**Actors**	**System**
	Incorrect data patterns generation.	Alert the user it is not sufficient data.

Use Case Name	Generate the sleep patterns	
Alternate flow	**Actors**	**System**
	-	-
Exclusions	Data correction option	
Post Conditions	Sleep patterns are shown to the user and stored in the database.	

Use Case Name	Update the data	
Use Case ID	UC-006	
Description	Users can update their old data	
Priority	Medium	
Primary Actor	User	
Supporting Actors	Developers, wearable devices	
Pre-Conditions	The user should enter his data first.	
Trigger	Click the Update button	
Main flow	**Actors**	**System**
	The user updates the data	Updates the data on the database

Use Case Name	Update the data	
Exception flow	**Actors**	**System**
	-	-
Alternate flow	**Actors**	**System**
	-	-
Exclusions	**History data modification**	
Post Conditions	**Data updated in the database.**	

Use Case Name	User Authentication	
Use Case ID	UC-007	
Description	Authenticate user to access the web app	
Priority	High	
Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	The user should register an account and give the validated details(email).	
Trigger	The user clicks the signup button	
Main flow	**Actors**	**System**

Use Case Name	User Authentication	
	The user enters his data	Validate the data and stored in the database
Exception flow	**Actors**	**System**
	User enter invalid data	Display a error
Alternate flow	**Actors**	**System**
	-	-
Exclusions	Register by social media or third-party platform.	
Post Conditions	The user creates the account after the user goes to the home page.	

4.7 Functional Requirements (with prioritization)

Explanation of prioritization.

1. Critical- The requirements that are critically needed in the successful completion
2. Desirable – The requirements that can add value, but are not required immediately
3. Luxury – The requirements that would add luxury to the system

Requirements list	**Priority Level**	**Description**

FR1	User data input system	Critical	Allow users to input them data (e.g. user's age, average sleep time, average study hours, caffeine intake etc.)
FR2	Sleep quality scoring	Critical	Provide sleep quality score based in user inputs.
FR3	Personalized recommendation	Critical	Offer advice to each user to improve their sleep quality score.
FR4	Goal setting and scheduling	Desirable	User can edit the sleep their goals and sleep schedules.
FR5	Sleep history and progress tracking	Desirable	User can get idea about his sleep history and he can know about his sleep quality is improved or not.
FR6	Data Analytics Dashboard	Desirable	The user can understand his sleep patterns.
FR7	Notification system	Luxury	This web app includes the notification for the bedtime and wakeup reminders.

FR8	Customizing and Language Support	Luxury	Including the three main languages in SriLanka (sinhala,tamil, and english)for this app. And user can access to customize his screen to dark mode or light mode.

4.8 Non- Functional Requirements

Requirements list		**Priority Level**	**Description**
NFR1	performance	Critical	When the user gives the data to the form then within 2 seconds or minimum time, predict the sleep quality.
NFR2	Security	Critical	All user's data must be protected during the transitions and the storage should protected.
NFR3	Scalability	Luxury	The web app can be handled by 500+ users without any issues.
NFR4	Usability	Critical	The user can accessible to the app by using minimum technical knowledge.

4.9 Chapter Summary

This chapter primary focus was on identifying the project stakeholders, understanding their viewpoints, selecting the requirement elicitation techniques, and identifying the use case diagrams, and getting to know about functional and non-functional requirements. Functional and non-functional requirements categorized based on the project's objectives. Stakeholders were shown by using Onion model, and give a viewpoint on their roles. And use case diagrams shows the web application core functionalities.

Chapter 5: Social, Legal, Ethical and Professional Issues

5.1 Chapter Overview

This section analyzes sociocultural, legal, moral, and professional aspects raised by the activities that involve the making and implementation of an artificial intelligence web-based application that enables forecasting, as well as the analysis of the sleeping patterns of university students. This project has implication for the collection of sensitive information from the users, predictive analysis-based techniques, and giving personal health advice. All these issues raised several ethical and legal concerns. In the next parts, these issues are addressed and the way to mitigate them in order to carry out responsible and law-abiding development are explained.

5.2 Dataset Ethical Clearance

Dataset source and description

The dataset used for this project, “Student sleep patterns”, was obtained from the Kaggle website. It includes data on ‘university students’ sleep duration, screen time, caffeine intake, physical activity, and student sleep quality.

- Dataset creator: Arsalan Jamal
- Dataset License: CC BY-SA 4.0 (Creative Commons Attribution-ShareAlike 4.0 International License)((*Deed - Attribution-NonCommercial-ShareAlike 4.0 International - Creative Commons*, n.d.))
- Dataset link: https://www.kaggle.com/datasets/arsalanjamal002/student-sleep-patterns

Additionally, collected the data from the IIT University and Sri Jayawardhanapura University students anonymously by the Google form.

5.3 SLEP Issues and Mitigation

5.3.1Overview

SLEP issues stand for social, Legal, Ethical, and Professional issues. This topic also discusses the sleep quality prediction classification domain’s ethical practice requirement and how to apply collected data without violating any of ethical guidelines and rules.

5.3.2 Social issues

This section discusses how the web app impacts to the society.

- This app is use for the improve the student's sleep quality and get better productivity. But some students can get thee anxiety about the sleep matrix.
- The project does not make for the multi-languages support.

5.3.3 Legal issues

This section discusses the project and compiles all relevant laws and regulations.

- The data privacy laws. Like GDPR, HIPAA, or CCPA (Plurilock Security, Inc., 2024). user's sensitive information should be have the privacy.
- Assuring transparency of the system works and user data is handled.
- Intellectual property legitimacy and authorization with respect to software, dataset, or API.

5.3.4 Ethical issues

This section discusses project ethical issues. Such as Protecting the user's privacy and securing the data. There are no fake references or details in the research effort.

5.3.5 Professional issues

From the beginning to end research was conducted in professional manner. Research work was evaluated by a team of domain and technological experts in every step. Prototyping will done under confidentiality and protected using all the latest security measures to prevent any unauthorized access to it.

5.4 Chapter Summary

This chapter explains the SLEP issues in the project and how these challenges were solved through ethical practices, professional standards, etc.Such initiatives guarantee the fulfillment of the project's objectives without compromising the rights and privacy of its users.

Chapter 6: System Architecture and Design

6.1. Chapter Overview

The system architecture and design are covered in this chapter. Author made some important choices when designing the system. This chapter makes these choices

very clearly. Author used information from the chapter on system requirement specifications to make these important design decisions. This chapter covered other potential design choices.

6.2. System Architecture Design

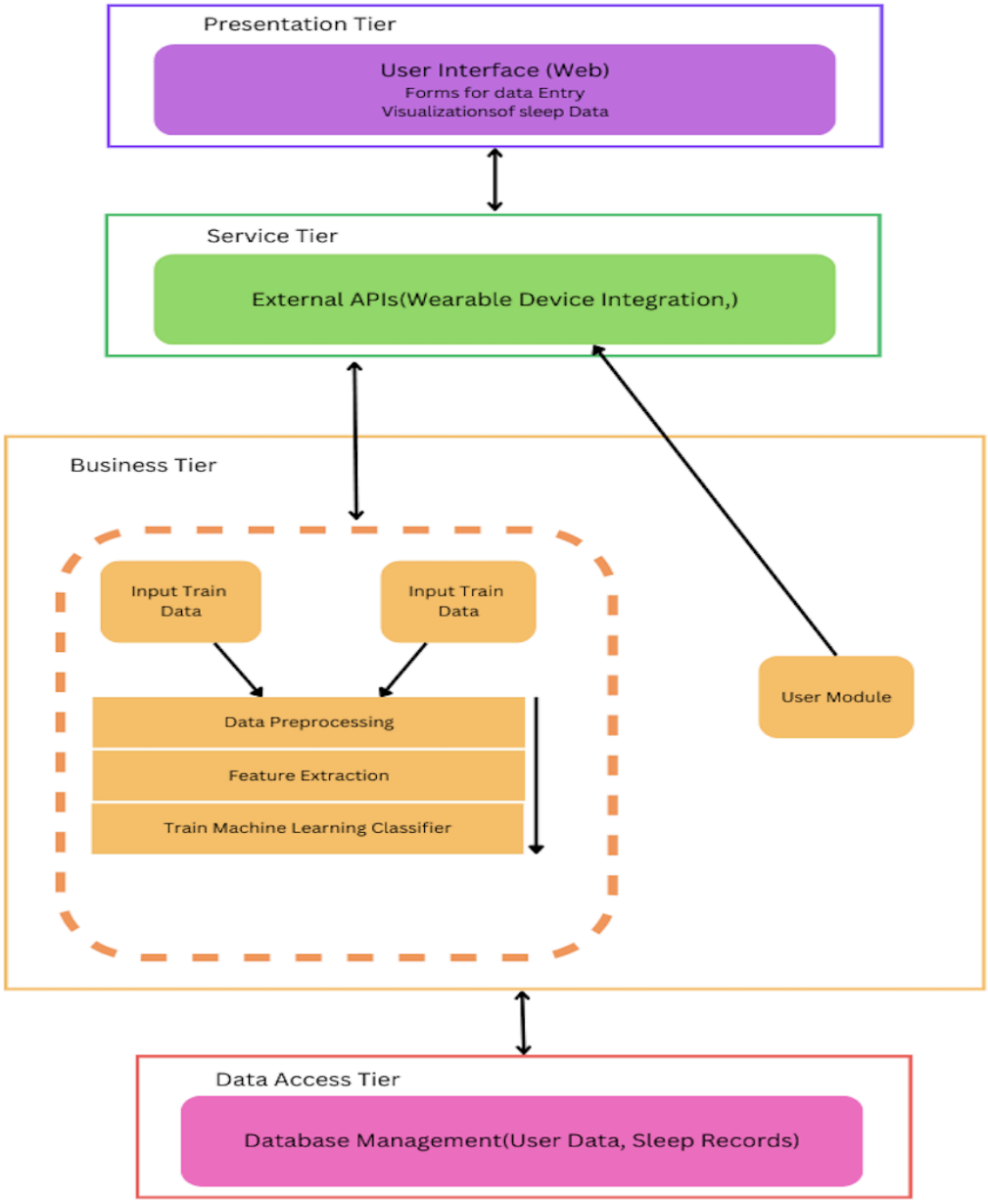


Figure 3: System Architecture Design

According to the tiered architecture diagram purposed system architecture consist of four layers. These layers are presentation tier, service tier, business tier and data access tier. Below contend explains each and every tier functionality and use.

6.2.1 Presentation tier

The web application user interface is designed to provide an intuitive engaging interface for user. University students enter their inputs like sleep duration, average study hours, etc. then view their sleep quality prediction and interact with tools like user sleep journals, and analytics dashboards.

6.2.2 Service tier

The Service Tier is the bridge for the presentation tier and the business tier. This layer handles the API requests and responses. When a user enters his data request and the service tier processes these API calls and retrieves the from the business tier and sends the data for the business tier.

6.2.3 Business tier

The business tier is the tier that holds all the backend logic. It contains the prediction and its training weight According to author's proposed system, the business tier contains data preprocessing and feature extraction functionalities too. Well-structured business tier shows the good coordination of business components of the system.

6.2.4 Data access tier

This tier stores the student sleep patterns dataset and all other bits of data that the system needs. Due to the nature of the project data is a crucial part of this system. Accuracy of data and validation are required for the system to function perfectly.

6.3. System Design

6.3.1.Class Diagram/Data Flow Diagram (DFD)

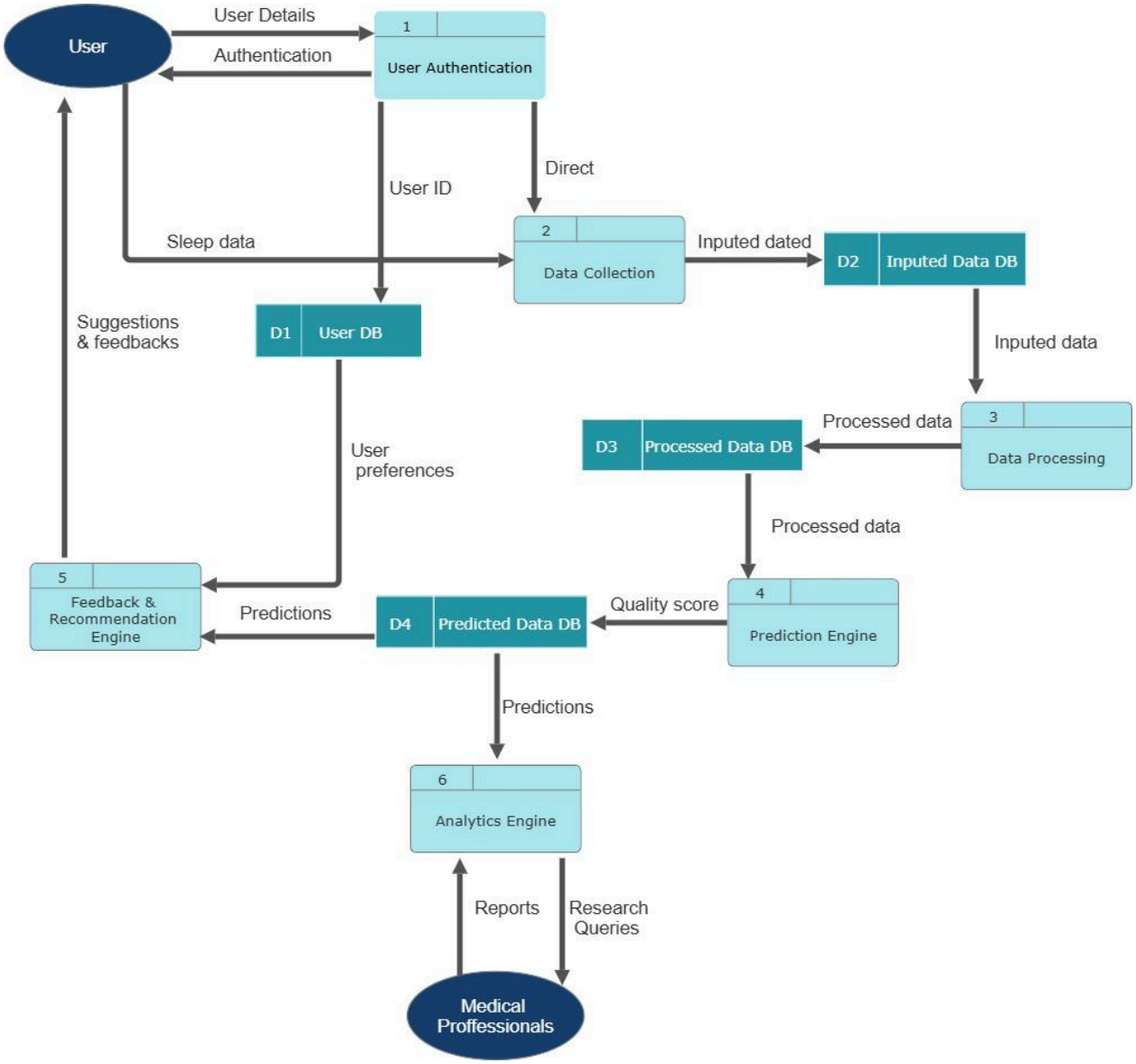


Figure 4: class diagram

6.3.2.Sequence Diagram

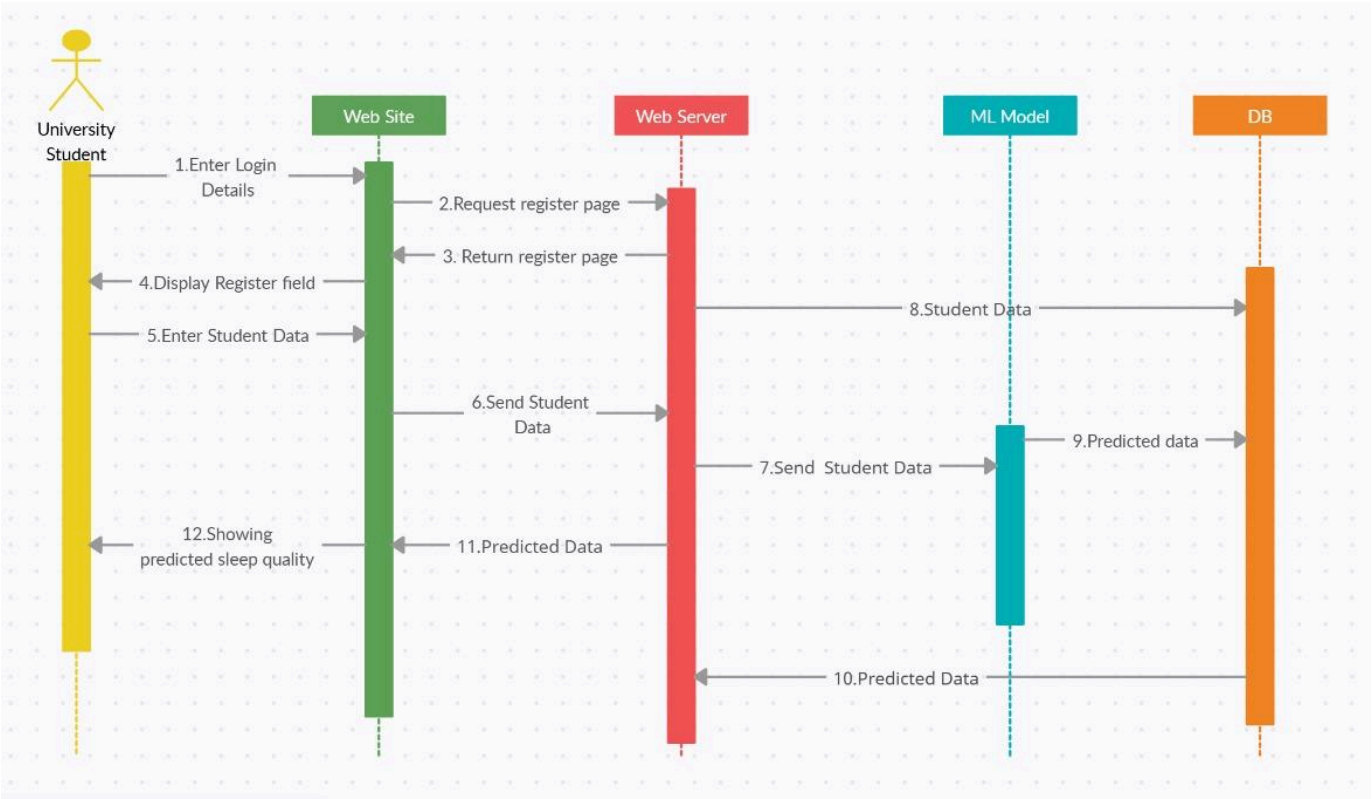


Figure 5:sequence diagram

6.3.3.UI Design and Mockups

Figure 5: UI design diagram

6.3.4.Activity Diagram

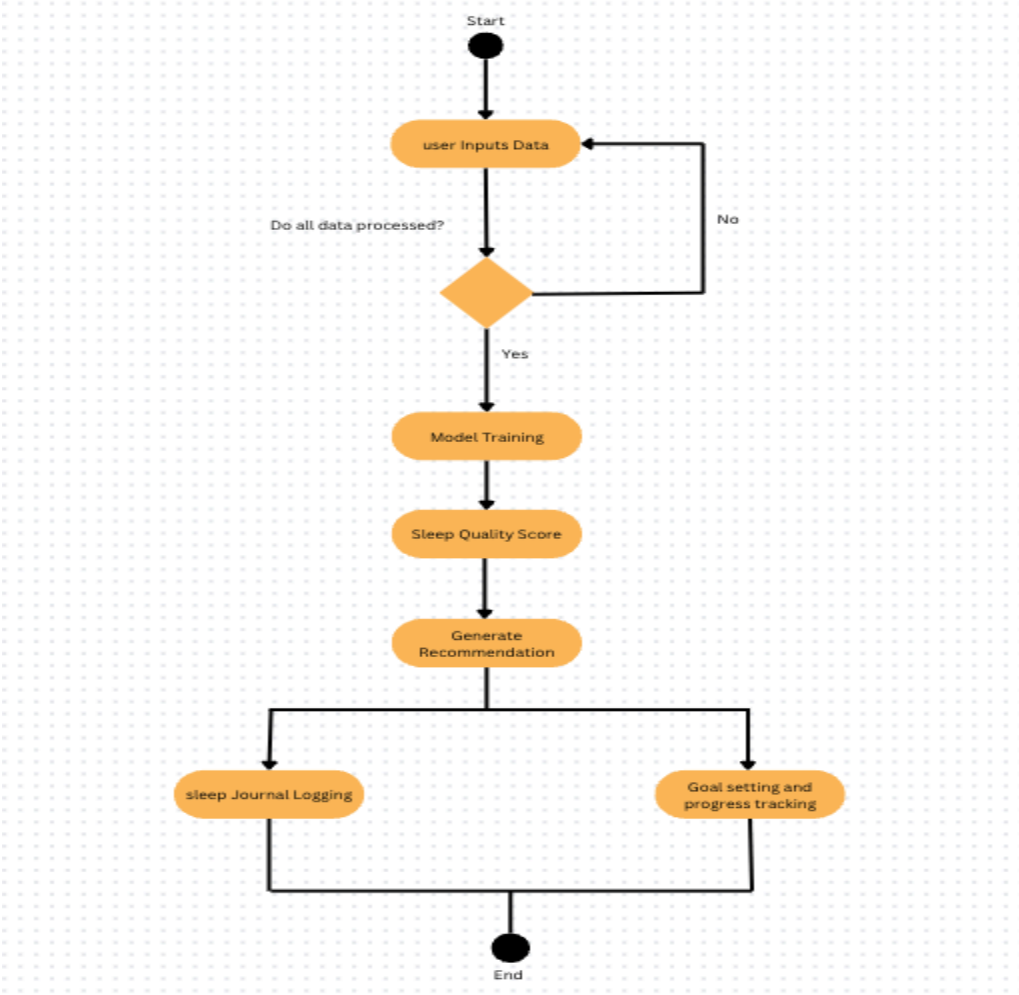


Figure 7: Activity diagram

6.4. Chapter Summary

This chapter covers design objectives, system architecture design, and system designs (class flow diagrams, sequence diagrams, UI Design, Activity diagrams) The structural design forms the basis of the systems architectural design, ensuring that both the functional and non-functional requirements are met in relation to the needs of university students.

References

1. Plurilock Security, Inc. (2024, May 31). *How GDPR, CCPA, HIPAA, and other data privacy standards safeguard our digital lives*. Plurilock. https://plurilock.com/blog/how-gdpr-ccpa-hipaa-and-other-data-privacy-standards-safeguard-our-digital-lives/
2. *Deed - Attribution-NonCommercial-ShareAlike 4.0 International - Creative Commons*. (n.d.). https://creativecommons.org/licenses/by-nc-sa/4.0/
3. *Student sleep patterns*. (2024, October 14). Kaggle. https://www.kaggle.com/datasets/arsalanjamal002/student-sleep-patterns
4. *Sleepless nights? Let's fix that!* (n.d.). Google Docs. https://docs.google.com/forms/d/1aVbwY88AX7CctMUQ50c1M88WXA6HwIsU-Linr59j-aQ/edit